EMPIRICAL STUDY OF WACC

Weighted Average Cost of Capital Analysis

Higher WACC → Higher Risk → Higher Required Return

Prepared By
Kanishk Chawla

WHAT IS WACC?

Weighted Average Cost of Capital

$$WACC = Ke \times We + Kd \times (1-t) \times Wd$$

Ke = Cost of Equity

Kd = Cost of Debt

We = Weight of Equity

Wd = Weight of Debt

t = Tax Rate

Cost of Equity (Ke)

Computed via CAPM: Rf + β × ERP
Using Daily, Weekly & Monthly Beta frequencies
Daily 1Y , Weekly 3Y and Monthly 5Y Historical Data used to calculate beta

Cost of Debt (Kd)

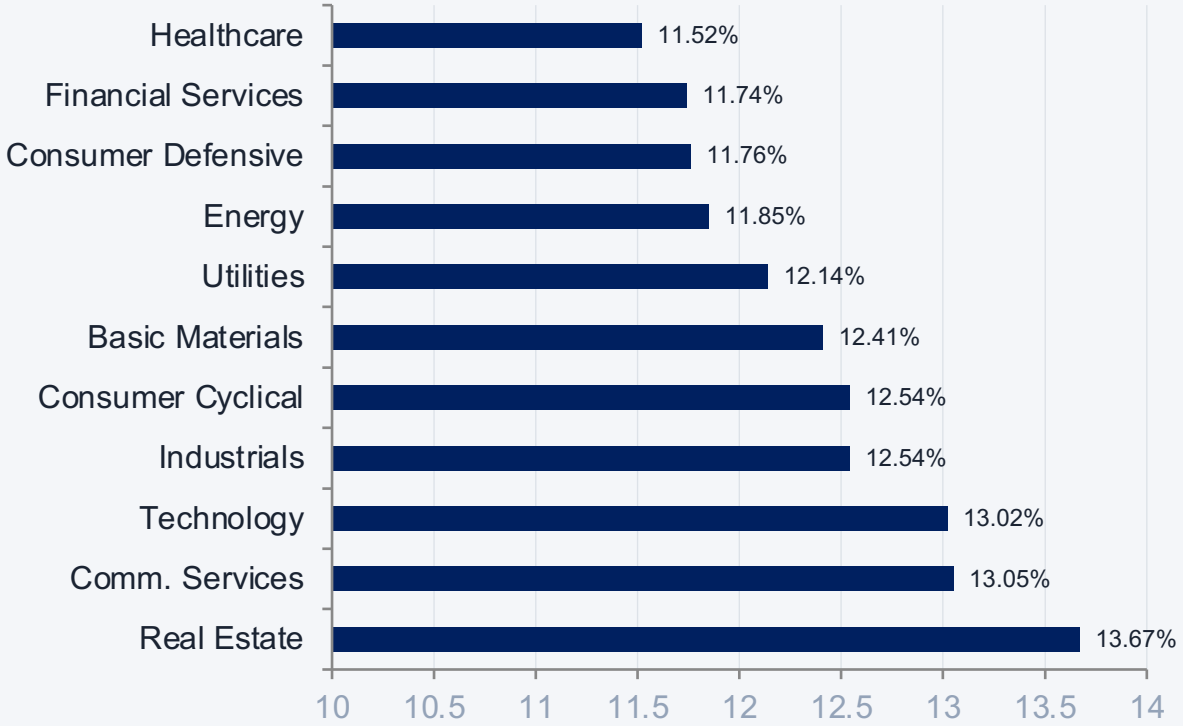
Derived from Interest Coverage Ratio → Credit Rating → Default Spread lookup table (Damodaran method)

Capital Weights

Market Value of Equity vs Market Value of Debt determines the proportion each component contributes

WACC ACROSS SECTORS

Average Daily WACC by Sector — NSE Listed Companies



Real Estate & Tech sectors carry the highest required return, reflecting greater operational and market risk.

Highest Risk

Real Estate

13.67%

Lowest Risk

Healthcare

11.52%

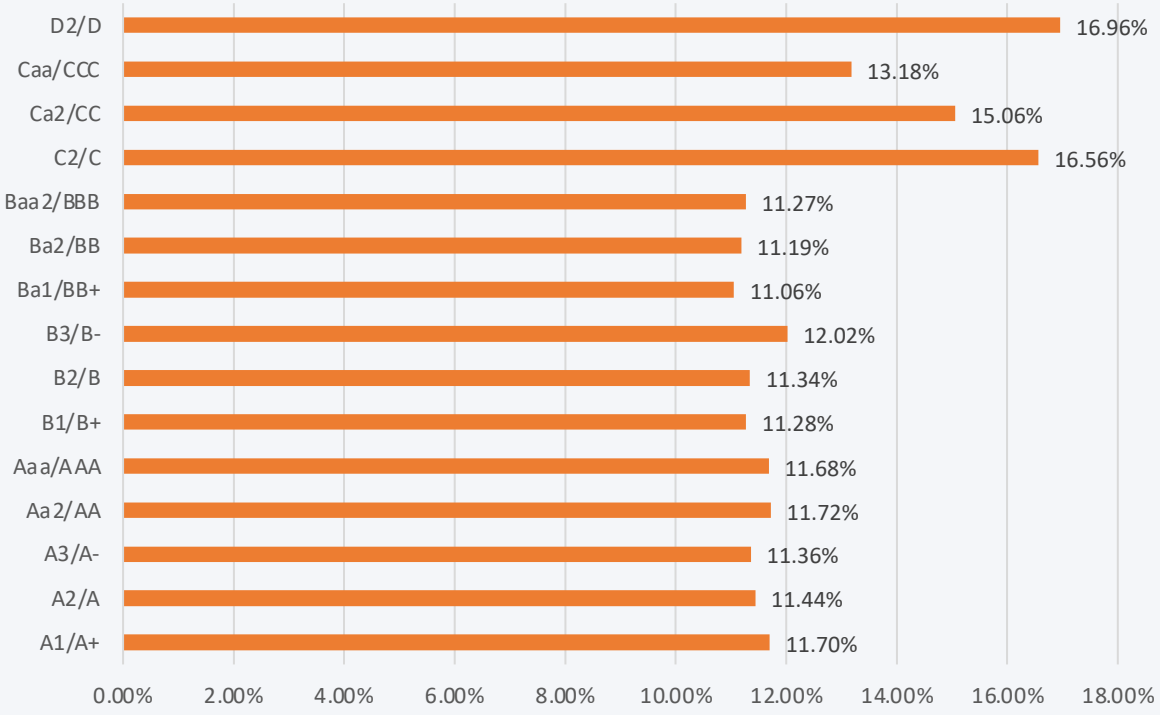
Spread

2.15 ppts

across sectors

WACC BY CREDIT RATING

Average Daily WACC — Higher Credit Risk = Higher WACC



Investment Grade (AAA–BBB)

Non-Investment (BB+–BB)

Speculative (B+–B-)

Distressed (CCC–D)

KEY FINDING

D2/D

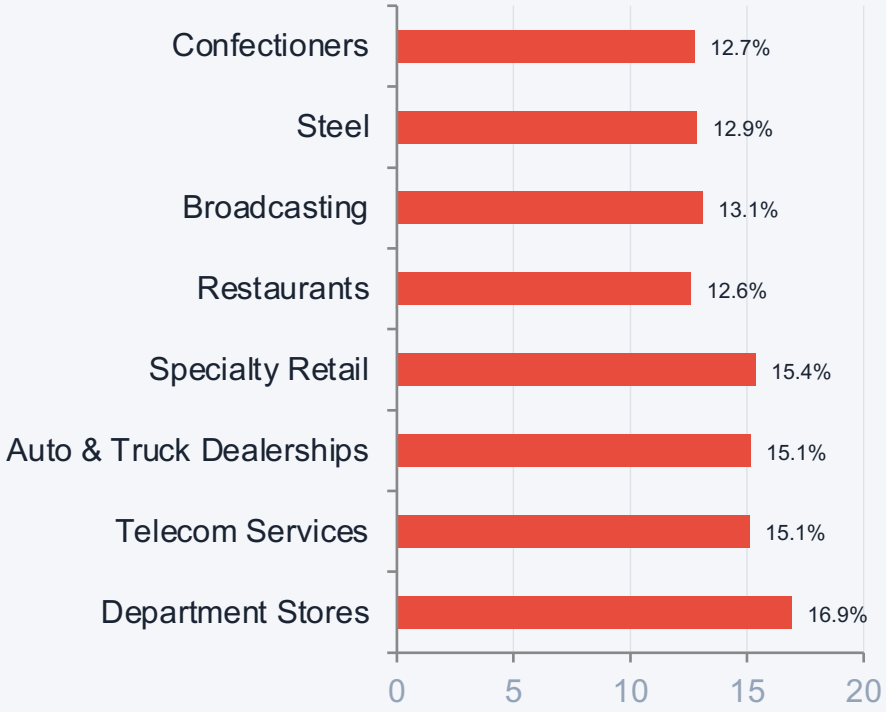
16.96%

Highest WACC

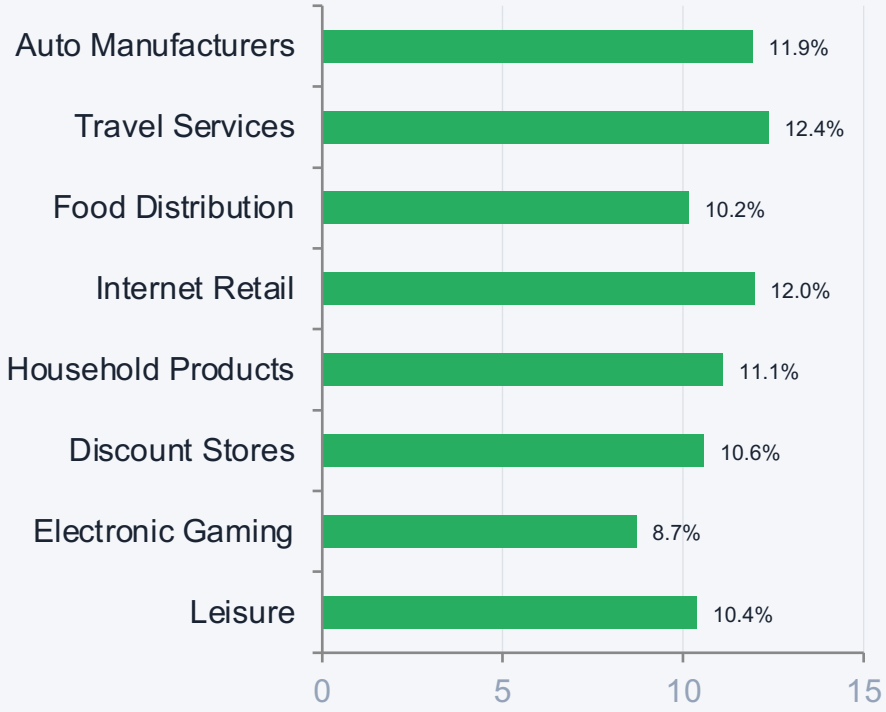
WACC BY INDUSTRY

Median Weekly WACC by Industry — Top & Bottom

HIGH RISK INDUSTRIES



LOW RISK INDUSTRIES



BETA & COST OF EQUITY

CAPM Framework: Ke = Rf + β × ERP | Rf = 6.67% | ERP = 5.23%

Daily Beta

Range: ~0.0 – 2.2

Avg Ke ≈ 12.4%

Captures short-term price volatility. Higher daily beta reflects greater sensitivity to market moves.

Weekly Beta

Range: ~0.0 – 2.1

Avg Ke ≈ 13.4%

Smoothed over weekly price movements. Generally closer to fundamental business risk.

Monthly Beta

Range: –1.0 – 2.2

Avg Ke ≈ 12.6%

Long-term systematic risk. Negative betas possible (counter-cyclical). Highest variance across firms.

β > 1: More volatile than market | β = 1: Moves with market | β < 1: Less volatile | β < 0: Counter-cyclical

KEY EMPIRICAL FINDINGS

01 Credit Rating is the Strongest WACC Driver

D2/D rated firms show WACC of 16.96% which higher than A1/A+ rated firms (11.7%). Credit risk dramatically dominates equity risk in determining capital costs.

02 Real Estate Sector Bears Highest Sectoral Cost

At 13.67% average daily WACC, Real Estate is the most capital-expensive sector. Healthcare at 11.52% offers the most stable, environment.

03 Beta Frequency Affects WACC Significantly

Weekly and monthly betas show markedly different cost of equity vs daily beta confirming that measurement frequency materially impacts empirical estimates.

04 Speculative Grade Firms Face > Cost vs Investment Grade

The jump from investment grade (BBB: ~11.27%) to speculative (B-: ~16.96%) is not linear distress risk creates exponential increases in required returns.

WACC = Risk Mirror | Higher Required Return = Signal of Elevated Risk

CONCLUSION

This empirical study demonstrates that WACC is not merely a financial formula it is a composite signal of market risk, credit quality, and sectoral dynamics.

- Credit ratings are the dominant driver — distressed firms face prohibitive capital costs
- Sectoral differences in WACC (~200 bps spread) reflect structural risk profiles
- Beta frequency selection materially alters the empirical cost of equity
- Investors should demand higher returns from high-WACC firms as compensation for risk

"In capital markets, WACC is the price of risk and risk always has a price."